Modulus Functions & the Factor/Remainder Theorem

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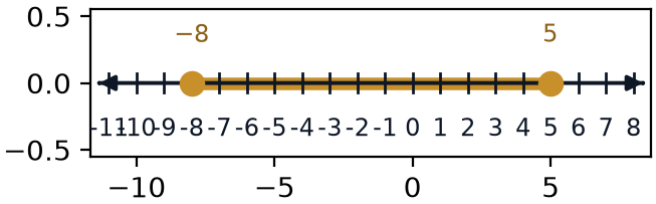
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = \frac{ax^2 + bx + c}{dx + e}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 3$. The graph also passes through the points $(0, 0)$ and $(1, -3)$. Find the value of $\frac{a}{d}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where $a, b, c, d \in \mathbb{R}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 3$ and a horizontal asymptote at $y = 2$. The function passes through the point $(1, -8)$. If $f'(x)$ has a critical point at $x = 5$, find the value of d.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Vectors & 3D Geometry · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

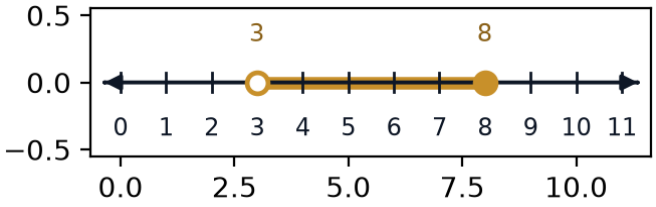
[Statistics & Probability (HL) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Calculus (HL) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

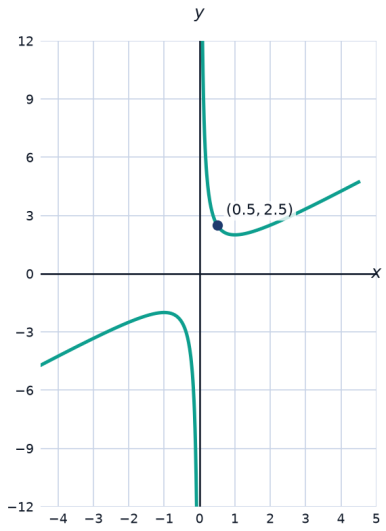
Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t) = 120 - 4t$ liters/hour and leaks out at a rate $L(t) = 3t$ liters/hour for $0 \le t \le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x) = x + \frac{1}{x}$ on $[1, 4]$, and find all values of c that satisfy the theorem.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · Medium Find the area of the triangle with vertices $A(1, 1, 0)$, $B(3, 0, 2)$, and $C(0, 2, 4)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2+y^2}\,dA$ where D is the semi-annular region $\{(x, y) : 1 \le x^2 + y^2 \le 4, y \ge 0\}$.

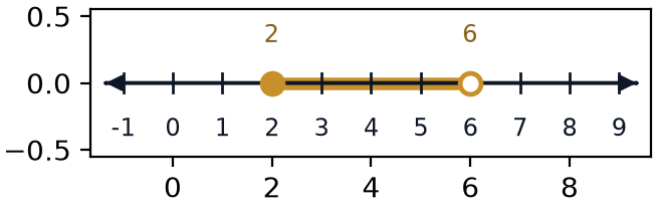
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2+5x-3}{x^2+4x+3}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.04 Partial Fractions

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \ge 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Functions g and h are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x-2}$ for $xeq2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi}\int_y^{\pi} \frac{\sin x}{x}\,dx\,dy$

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Level 2 · Medium Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the counterclockwise circle $x^2 + y^2 = 4$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why can $|x - 7| = -3$ have no real solutions?

a. The absolute value of any real expression is non-negative ($|u| \geq 0$), so it cannot equal a negative number

b. Because x must be greater than 7

c. Because -3 is not a prime number

d. The equation has complex solutions only

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the inequality $|2x - 3| < 7$.

a. $-2 < x < 5$

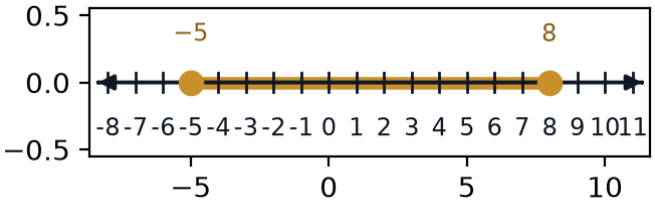
b. $x < 5$

c. $x < -2$ or $x > 5$

d. $-5 < x < 5$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which interval notation represents the solution to $|x - a| \leq \varepsilon$ where $\varepsilon > 0$?



a. $[a - \varepsilon, a + \varepsilon]$

b. $(a - \varepsilon, a + \varepsilon)$

c. $[-\varepsilon, \varepsilon]$

d. $[a, a + \varepsilon]$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When $P(x) = 2x^3 - 3x^2 + ax - 5$ is divided by $(x - 2)$, the remainder is 11. Find a.

a. 6

b. 4

c. -6

d. 8

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $(x+3)$ is a factor of $Q(x) = x^3 + kx^2 - 7x + 6$, find k.

a. 2

b. -2

c. 3

d. -1

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A polynomial $f(x)$ leaves remainder 3 when divided by $(x-1)$ and remainder 7 when divided by $(x-3)$. What is the linear remainder when $f(x)$ is divided by $(x-1)(x-3)$?

a. $2x+1$

b. $2x-1$

c. $x+2$

d. $4x-1$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How is the graph of $y = |f(x)|$ obtained from the graph of $y = f(x)$?

a. Any part of the graph lying below the x-axis is reflected vertically across the x-axis into the upper half-plane

b. The left half of the graph is discarded and replaced by reflection of the right half

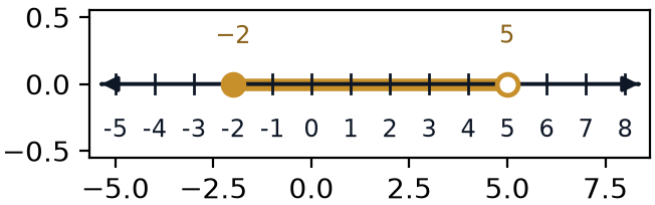
c. The graph is shifted 1 unit upward

d. The graph is reflected across the y-axis

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How is the graph of $y = f(|x|)$ obtained from $y = f(x)$?



a. Discard the portion for $x < 0$, and reflect the portion for $x \geq 0$ across the y-axis to form an even function

b. Reflect the entire graph across the x-axis

c. Reflect the graph across $y = x$

d. Shift the graph to the right by $|x|$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the vertex (sharp turn) of the function $g(x) = 3|x - 4| - 5$?

a. $(4, -5)$

b. $(-4, -5)$

c. $(4, 5)$

d. $(0, -5)$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve $|x + 1| = |-3x - 1|$. Give the largest solution.

a. -2

b. 4

c. 3

d. 0

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Factor $P(x) = x^3 - x^2 - 4x + 4$ completely. Find the largest root.

a. 2

b. 1

c. 4

d. 3

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the remainder when $P(x) = 2x^2 + 3x - 4$ is divided by $(x - 4)$.

a. 40

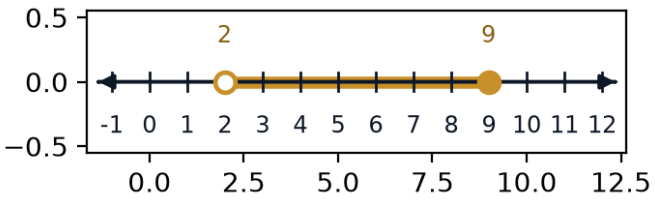
b. 32

c. 29

d. 47

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve $|x - 6| = 2x - 9$, given the solution is positive.



a. 6

b. 7

c. 5

d. 4

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Is $f(x) = 2x^3$ odd, even, or neither?

a. Odd

b. Neither

c. Both odd and even

d. Even

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given that $(x - 2)$ is a factor of $P(x) = 3x^2 - 6 + kx$, find k.

a. -7

b. -4

c. -3

d. -2

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate $\tan^{-1}(1)$ in degrees.

a. 67°

b. 22°

c. 69°

d. 45°

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given that f is an even function and $f(9) = 10$, find $f(-9)$.

a. 4

b. 10

c. 13

d. 14

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Given that f is an odd function and $f(6) = -5$, find $f(-6)$.

a. 8

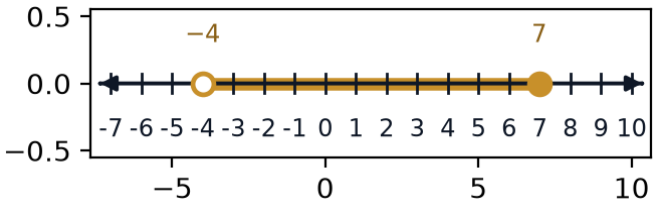
b. 1

c. 2

d. 5

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

If $f(x) = \frac{4}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).



a. 2

b. 1

c. 3

d. 4

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. 27

d. 26

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where a, b, c, and d are constants. The graph of $f(x)$ has a vertical asymptote at $x = 3$ and an oblique asymptote $y = 2x + 1$. Given that $f(1) = -4$ and $f(2) = -6$, find the value of a.

a. -2

b. 2

c. -4

d. 4

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.02 The Poisson Distribution

Consider the tetrahedron OABC with vertices O(0, 0, 0), A(2, 1, 3), B(1, 2, -1), and C(k, k+1, k-1). Given that the volume of the tetrahedron is 7 cubic units, find the possible values of k.

a. k = 1 or k = 4

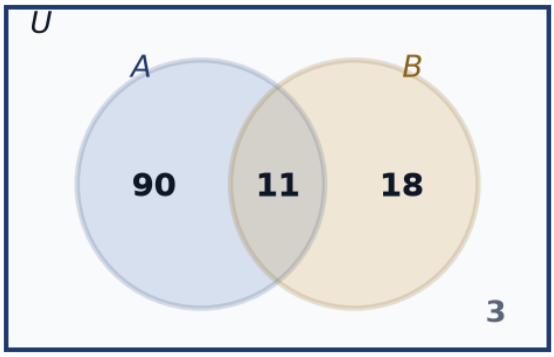
b. k = 2 or k = 3

c. k = 3 or k = 4

d. k = 4 or k = 5

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Further Integration Techniques

A school offers two optional subjects: Art and Music. The number of students enrolled in each subject and the number of students enrolled in both is shown in the Venn diagram below. The total number of students in the school is 120. Let A be the set of students enrolled in Art and M be the set of students enrolled in Music. Given that $n(A \cup M) = 90$, find the number of students enrolled in neither Art nor Music.



a. 20

b. 30

c. 40

d. 10

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Differential Equations

The following bar chart shows the number of students who chose each of four different subjects in a school. Let N be the total number of students surveyed. Find the value of N.

a. 90

b. 100

c. 96

d. 104

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Consider the points A(2, 3) and B(8, 7) in the Cartesian plane. The vector $\vec{AB}$ can be represented as $\mathbf{v}$. Find the components of $\mathbf{v}$.

a. (6, 4)

b. (10, 10)

c. (-6, -4)

d. (4, 6)

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